分子文本表示实验结果记录

PDF 使用文本布局导出 与 DOCX 分页可能不同

分子文本表示实验结果记录

依据保存的真实响应与评分文件生成 需人工核对后汇报

当前状态为计划中的模型调用已全部保存。预定 270 条请求，API 成功 270 条，能解析最终答案 269 条。调用成功不代表答对，也不代表已完整复现论文。

本次执行的是基于作者公开数据、经审计后适配的复现协议。B5 标签规则、分子对成员表示、选项顺序、生成样本数及停止序列等与公开脚本存在记录的差异。论文与公开 B9/B10 数据源说明也不一致。

原论文研究同一个分子在不同文本表示下的 LLM 表现是否改变。RDKit 用于结构解析、格式转换及标准答案计算；模型只获得任务提示词，标准答案不发送给模型。

模型与证据

结果目录 C:\Users\Asus\Desktop\molrepbench\molrepbench_reproduction\results\deepseek-flash_off_smoke_run3

结果类型 real_model_inference

模型 deepseek-flash

推理标签 off 实际开关须核对后端

原始回答 raw.jsonl 逐题评分 scored.jsonl 汇总 summary.csv 状态 evaluation_status.json

分任务与分表示成绩

空值表示未完成、无法计算或解码未评估。连续性质仅在成功解析的数字上计算相关性，同时显示覆盖率。B2 缺答案时主 Macro-F1 留空。

可逐条追溯的案例

案例 1

B3:439741:logp:canonical_smiles

实际 prompt
Estimate the LogP (partition coefficient) of the following molecule.
Molecule: Nc1ncnc2c1ncn2C1OC(COP(=O)(O)OP(=O)(O)OCC(O)C(=O)O)C(O)C1O
Please reason step by step, and put your final answer (a number) within \boxed{}.
Example format: After reasoning, conclude with: The answer is \boxed{3.5}

模型原始回答
This molecule is a nucleotide analog: specifically, it looks like an adenine nucleotide derivative with a modified sugar and a diphosphate linked to a glyceric acid-like moiety. Let ' s break it down.
Step 1 – Identify key fragments and their contributions
1. **Adenine base** (Nc1ncnc2c1ncn2 –): aromatic, quite polar, LogP contribution roughly − 0.1 to 0.1. Adenine alone has LogP ≈ − 0.1.
2. **Sugar ring** (C1OC(...)C(O)C1O): a ribose-like tetrahydrofuran with two hydroxyls. Very hydrophilic, contributes significantly negative LogP (each OH ≈ − 0.7 to − 1.0 in LogP terms). The sugar itself is very polar.
3. **Phosphate groups**: Two phosphate groups (P(=O)(O)⋯) — ionizable, highly hydrophilic. At physiological pH they ' re ionized; in neutral form for LogP calculation, still very polar. Each phosphate can contribute − 1.5 to − 2 or more to LogP.
4. **Glyceric acid-like end** (– OCC(O)C(=O)O): contains a carboxylic acid and a hydroxyl — very polar, ionizable, LogP contribution strongly negative.
Step 2 – General estimation approach

Such a molecule is large and has multiple H-bond donors/acceptors, several charges possible. Experimentally, nucleotides and sugar phosphates have very low or negative LogP (often −2 to −5). This molecule has:
- Adenine (slightly hydrophobic but small)
- Sugar (hydrophilic)
- Two phosphates (very hydrophilic)
- Carboxylic acid (hydrophilic)
The sum of hydrophilic parts far outweighs the small aromatic part.
Step 3 – Rough numerical sum
- Adenine: ~0
- Sugar with 2 OH: ~ −1.5
- Two phosphates: ~ −3 to −4
- Carboxylic acid + OH: ~ −1.5
Total: around −6, but such fragments in LogP estimation often compress toward less extreme values due to intramolecular interactions. Many phosphate-containing nucleotides have LogP in the −3 to −5 range.
Step 4 – Comparison to similar molecules
ATP has LogP around −5.5 (varies by source, often −5 to −6). This molecule is similar but with one less phosphate and an extra carboxylic acid — probably similar or slightly less negative. NAD has LogP around −4 to −5.
Given it's a dinucleotide-like with two phosphates and a carboxylate, I'd estimate around −4 to −5. Let's settle on −4.5 as a reasonable middle estimate.
Final answer:
The answer is \boxed{-4.5}

参考答案 -2.276199999999999 解析结果 -4.5

案例 2

B3:439741:tpsa:canonical_smiles

实际 prompt
Estimate the Topological Polar Surface Area (TPSA) of the following molecule.
Molecule: Nc1ncnc2c1ncn2C1OC(COP(=O)(O)OP(=O)(O)OCC(O)C(=O)O)C(O)C1O
Please reason step by step, and put your final answer (a number) within \boxed{}.
Example format: After reasoning, conclude with: The answer is \boxed{3.5}

模型原始回答
Let's break down the molecule and estimate TPSA.
Step 1 – Identify the molecule
The given SMILES is:
`Nc1ncnc2c1ncn2C1OC(COP(=O)(O)OP(=O)(O)OCC(O)C(=O)O)C(O)C1O`
This is an adenine nucleotide analog with a diphosphate group and a ribose-like sugar, attached to something like a glycerate-like moiety via pyrophosphate. Actually, it looks like **adenosine diphosphate ribose** derivative? Wait, the sugar is not ribose—it's a cyclopentane? Let's parse:
- `Nc1ncnc2c1ncn2` = adenine base (with NH2 on C6)
- Attached to `C1OC(...)C(O)C1O` — this is a furanose ring? Actually it's a tetrahydrofuran-type sugar (like ribose) but written in a way that shows substituents: C1 (anomeric) attached to adenine N9, then O in ring, then C with CH2-O-P..., then C(OH), then C(OH) closing ring back to C1. So yes, it's a ribose.
- The 5'-O is connected to `P(=O)(O)OP(=O)(O)OCC(O)C(=O)O` — that's a diphosphate, then to `OCC(O)C(=O)O`, which is a 3-phosphoglycerate-like or glycerate? Actually `OCC(O)C(=O)O` is glyceric acid 3-O-? Wait, O-C-C(OH)-C(=O)O: 3 carbons: C1=O, C2-OH, C3-O-P. That's 3-phosphoglycerate fragment. So the molecule is **adenosine 5'-diphospho-3-phosphoglycerate**? Or maybe ADP-ribose-1''-phosphate? No, it's ADP attached to 3-phosphoglycerate? Possibly it's **ADP-ribose pyrophosphatase substrate**? It's actually **ADP-ribose 1''-phosphate**? Let's not worry about the name—just

count polar atoms.
Step 2 – List polar atoms and their environments
We'll count N, O, and attached H for TPSA.
Adenine part:
- NH2 (aromatic amine): N with 2 H → TPSA contribution ~26.02
- Aromatic N's in purine: 4 ring nitrogens (N1, N3, N7, N9). N9 is substituted (attached to sugar) so is aromatic N with no H (pyrrole-type? Actually N9 in adenine is glycosidic, so it's aromatic N with 3 bonds, no H → contributes ~4.93 each? Wait, need standard Ertl TPSA contributions.
- Let's recall Ertl's table:
- [nH] (pyrrole-type) ~ 15.79
- [n] (aromatic N no H) ~ 12.89 (e.g., pyridine N)
- [NH2] aromatic ~ 26.02
- [N] aliphatic amine ~ 3.24 (tertiary), etc.
But N9 here is aromatic, connected to 3 carbons? Actually N9 in purine is attached to C4, C8, and sugar C1'—so it's aromatic N with 3 bonds, no H → ~4.93? Wait, that value seems too low. Let me check: In Ertl's paper, aromatic N with 3 neighbors (pyrrole-type but substituted) ~ 4.93? No, that's for N in pyrrole with H? Actually pyrrole N-H is 15.79, pyrrole N substituted is 4.93. Yes, that's correct. Pyridine-type N is 12.89.
So:
Adenine:
- NH2: 26.02
- N1 (pyridine-type): 12.89
- N3 (pyridine-type): 12.89
- N7 (pyridine-type): 12.89
- N9 (pyrrole-type, substituted): 4.93
Sum for adenine N's = 26.02 + 12.89*3 + 4.93 = 26.02 + 38.67 + 4.93 = 69.62
Sugar part (ribose):
- Ring O: ether oxygen ~ 9.23
- Two OH groups (2',3') each ~ 20.23 → 40.46
- 5'-O is phosphate ester: that O is part of phosphate, we count in phosphate.
Sugar contributions: O(ring) 9.23 + 2*OH 40.46 = 49.69
Phosphate groups:
First phosphate: P(=O)(O)-O-
- P=O oxygen: 17.07? Wait, need to recall:
Ertl TPSA for phosphate: each P=O ~ 17.07? Actually O in P=O ~ 17.07, O in P-OH ~ 20.23? But here we have P(=O)(O)OP...
Let's use known fragments:
Phosphate monoester: R-O-P(=O)(OH)2:
O(ester) ~ 9.23, P=O ~ 17.07, two OH ~ 2*20.23=40.46 → total ~ 66.76? That seems high. Actually in Ertl's table, phosphate O's:
- O in P=O: 17.07
- O in P-OH: 20.23
- O in P-O-C: 9.23
But here we have diphosphate: R-O-P(=O)(OH)-O-P(=O)(OH)-O-R'
Let's count all O's in diphosphate bridge:
- O linking sugar to P1: 9.23
- P1=O: 17.07
- P1-OH: 20.23
- O bridging P1-O-P2: 9.23? Actually bridging O is P-O-P, TPSA ~ 9.23? I think bridging O in anhydride ~ 9.23? Possibly yes.
- P2=O: 17.07

- P2-OH: 20.23
- O linking P2 to glycerate: 9.23
Sum so far for diphosphate part: 9.23+17.07+20.23+9.23+17.07+20.23+9.23 = 102.29
Glycerate part: OCC(O)C(=O)O
- O linking to P: counted above (9.23)
- OH on C2: 20.23
- Carboxylic acid: C(=O)OH → O=C ~ 17.07? Actually carboxyl: =O ~ 17.07, -OH ~ 20.23 → total ~ 37.3? Wait, Ertl: COOH = 37.3
- So glycerate: OH (20.23) + COOH (37.3) = 57.53 (O linking to P already counted).
Step 3 – Sum everything
Adenine N’s: 69.62
Sugar: 49.69
Diphosphate + linkers: 102.29
Glycerate (OH + COOH): 57.53
Total = 69.62 + 49.69 = 119.31
119.31 + 102.29 = 221.60
221.60 + 57.53 = 279.13
That seems high but plausible for a molecule with many OH, phosphate, carboxylate.
Let’s check: Known TPSA for ATP is ~279? ATP TPSA is 279 Å² indeed. Our molecule has similar count: ATP has triphosphate, adenine, ribose. Our molecule has diphosphate + carboxylate + extra OH. ATP: adenine (69.62), ribose (49.69), triphosphate (~130?), total ~ 250? Actually ATP TPSA ~ 279. So our value ~279 is reasonable.
But our molecule: diphosphate (~102) + glycerate part (~57) vs ATP triphosphate (~130). ATP: 69.62+49.69+~160? No, triphosphate TPSA ~ 150? Let’s check ATP: 3 phosphates: each ~ 63? Actually ATP TPSA = 279. So 69.62+49.69=119.31, remainder 159.69 for triphosphate. Our diphosphate ~102 + glycerate extra polar ~57 = 159, matches. So total ~279.
Given slight variations, likely ~279.
Step 4 – Final check
Possible small differences: N9 in adenine sometimes counted differently, but 279 is a good estimate.

The answer is \boxed{279}

参考答案 279.13 解析结果 279.0

案例 3

B3:439741:hbd:canonical_smiles

实际 prompt
Estimate the number of hydrogen bond donors of the following molecule.
Molecule: Nc1ncnc2c1ncn2C1OC(COP(=O)(O)OP(=O)(O)OCC(O)C(=O)O)C(O)C1O
Please reason step by step, and put your final answer (a number) within \boxed{}.
Example format: After reasoning, conclude with: The answer is \boxed{3.5}

模型原始回答
Let's break this down step by step.
Step 1: Understanding the molecule
The given SMILES is:
`Nc1ncnc2c1ncn2C1OC(COP(=O)(O)OP(=O)(O)OCC(O)C(=O)O)C(O)C1O`
We can identify parts:
- It’s a nucleotide-like molecule: looks like an adenine base linked to a sugar, which is linked to a diphosphate, which is linked to another fragment containing a carboxylic acid.
- Actually, this looks like **adenosine diphosphate ribose** or something similar but with some differences: The base is adenine

(Nc1ncnc2c1ncn2⋯), linked to a ribose (C1OC(...)C(O)C1O).
- The phosphate chain: `COP(=O)(O)OP(=O)(O)OCC(O)C(=O)O` — a diphosphate linking to a 3-carbon chain with OH and COOH at the end. That's like ADP-ribose but with a modified terminal (maybe ADP-glycerate?).
Anyway, we need **hydrogen bond donors (HBD)** count.
Step 2: Definition of HBD
Hydrogen bond donors are typically N – H and O – H groups (where H is covalently bound to N or O). Each N – H or O – H counts as one donor (some definitions count NH2 as 2 donors, some as 1 group = 1 donor? Usually in cheminformatics, each hydrogen atom attached to N or O counts separately). We ' ll follow the standard: count each N – H and O – H as 1 donor.
Step 3: Counting N – H donors
The adenine ring:
- `Nc1ncnc2c1ncn2...` → The exocyclic amino group (NH2) on the purine: that ' s 2 N – H bonds → 2 donors.
The ring nitrogens: in aromatic rings, if they have no H (they don ' t here, they ' re either tertiary or imine-like without H), they don ' t count. So only the NH2 gives 2 donors.
No other N – H in molecule (no amide, no amine besides that). So N – H donors = 2.
Step 4: Counting O – H donors
Let ' s go through all oxygens:
1. Sugar ring: `C1OC(...)C(O)C1O` → two hydroxyls on the ribose: 2 O – H groups → 2 donors.
2. Phosphate groups: `P(=O)(O)` → each has one P – OH (the other oxygens are P=O or P – O – C/P – O – P). There are two phosphate groups: each has one OH → 2 donors.
3. The `OCC(O)C(=O)O` part: one OH on the carbon chain (`C(O)`) → 1 donor; the carboxylic acid `C(=O)O` has an OH → 1 donor.
4. Check if any other OH: the `COP(=O)(O)` — first phosphate has OH counted. The second phosphate `OP(=O)(O)OCC...` also has OH counted.
Let ' s recount systematically:
- Sugar: 2 OH groups (positions 2' and 3' of ribose).
- Phosphate 1: 1 OH.
- Phosphate 2: 1 OH.
- Glycerate-like part: 1 OH (secondary alcohol) + 1 COOH → 2 OH.
Total O – H = 2 (sugar) + 1 + 1 (phosphates) + 2 (other fragment) = 6 O – H donors.
Step 5: Total HBD
N – H: 2
O – H: 6
Total = 8.
But let ' s check for possible missing or extra:
- The Nc1ncnc2c1ncn2 — NH2 is 2 H ' s → 2 donors.
- Any other NH? No.
- Oxygens: Let's list all O atoms in SMILES:
`Nc1ncnc2c1ncn2C1OC(COP(=O)(O)OP(=O)(O)OCC(O)C(=O)O)C(O)C1O`
Oxygens:
1. Sugar ring O (ether) — no H.
2. Sugar OH (C(O)) — 1 H.
3. Sugar OH (C1O) — 1 H.
4. Phosphate: P(=O)(O) — 1 OH.
5. Phosphate: P(=O)(O) — 1 OH.
6. OCC(O)... — OH → 1 H.
7. C(=O)O — OH → 1 H.
8. Any others? The phosphate bridging O ' s: P – O – P, P – O – C — no H. The C=O in COOH — no H.
So yes: 6 O – H.

Step 6: Final count
2 (N – H) + 6 (O – H) = 8.
The answer is \boxed{8}

参考答案 7 解析结果 8.0

我可以据此解释的内容

我比较的是模型读取不同分子写法后的答题表现。输入格式的显式程度、语言线索和可保留的信息不同，因此需要同时检查任务、

汇报前我还需确认 IUPAC
输入审核、生成解码、FCD、配对置信区间及模型版本；原文裁判图和机制分析需另列完成情况。程序未自动写入任何“ 全文成功

尚需人工完成

逐项对照论文结果；解释真实失败案例；把实际模型、数据与协议偏差补齐；确认哪些结论被本次实验支持。请同时阅读随包的说

B1 all | canonical_smiles | 2 | 100.0% | 0.5000

B1 all | cml | 2 | 100.0% | 1.0000

B1 all | deepsmiles | 2 | 100.0% | 1.0000

B1 all | inchi | 2 | 100.0% | 1.0000

B1 all | isomeric_smiles | 2 | 100.0% | 0.5000

B1 all | iupac | 2 | 100.0% | 1.0000

B1 all | moljson | 2 | 100.0% | 1.0000

B1 all | randomized_smiles | 2 | 100.0% | 1.0000

B1 all | selfies | 2 | 100.0% | 0.0000

B2 all | canonical_smiles | 10 | 100.0% | 0.4000

B2 all | cml | 10 | 100.0% | 0.4000

B2 all | deepsmiles | 10 | 100.0% | 0.3333

B2 all | inchi | 10 | 100.0% | 0.4000

B2 all | isomeric_smiles | 10 | 100.0% | 0.4000

B2 all | iupac | 10 | 100.0% | 0.4000

B2 all | moljson | 10 | 100.0% | 0.4000

B2 all | randomized_smiles | 10 | 100.0% | 0.4000

B2 all | selfies | 10 | 100.0% | 0.1333

B3 hba | canonical_smiles | 2 | 100.0% | 0.0000

B3 hba | cml | 2 | 100.0% | 0.0000

B3 hba | deepsmiles | 2 | 100.0% | 0.0000

B3 hba | inchi | 2 | 100.0% | 0.5000

B3 hba | isomeric_smiles | 2 | 100.0% | 0.0000

B3 hba | iupac | 2 | 100.0% | 0.5000

B3 hba | moljson | 2 | 100.0% | 0.0000

B3 hba | randomized_smiles | 2 | 100.0% | 0.0000

B3 hba | selfies | 2 | 100.0% | 0.0000

B3 hbd | canonical_smiles | 2 | 100.0% | 0.5000

B3 hbd | cml | 2 | 100.0% | 0.5000

B3 hbd | deepsmiles | 2 | 100.0% | 0.0000

B3 hbd | inchi | 2 | 100.0% | 0.5000

B3 hbd | isomeric_smiles | 2 | 100.0% | 0.5000

B3 hbd | iupac | 2 | 100.0% | 0.5000

B3 hbd | moljson | 2 | 100.0% | 0.0000

B3 hbd | randomized_smiles | 2 | 100.0% | 0.5000

B3 hbd | selfies | 2 | 100.0% | 0.0000

B3 logp | canonical_smiles | 2 | 100.0% | 未评估

B3 logp | cml | 2 | 100.0% | 未评估

B3 logp | deepsmiles | 2 | 100.0% | 未评估

B3 logp | inchi | 2 | 100.0% | 未评估

B3 logp | isomeric_smiles | 2 | 100.0% | 未评估

B3 logp | iupac | 2 | 100.0% | 未评估

B3 logp | moljson | 2 | 100.0% | 未评估

B3 logp | randomized_smiles | 2 | 100.0% | 未评估

B3 logp | selfies | 2 | 100.0% | 未评估

B3 tpsa | canonical_smiles | 2 | 100.0% | 未评估

B3 tpsa | cml | 2 | 100.0% | 未评估

B3 tpsa | deepsmiles | 2 | 100.0% | 未评估

B3 tpsa | inchi | 2 | 100.0% | 未评估

B3 tpsa | isomeric_smiles | 2 | 100.0% | 未评估

B3 tpsa | iupac | 2 | 100.0% | 未评估

B3 tpsa | moljson | 2 | 100.0% | 未评估

B3 tpsa | randomized_smiles | 2 | 50.0% | 未评估

B3 tpsa | selfies | 2 | 100.0% | 未评估

B4 all | canonical_smiles | 2 | 100.0% | 1.0000

B4 all | cml | 2 | 100.0% | 1.0000

B4 all | deepsmiles | 2 | 100.0% | 1.0000

B4 all | inchi | 2 | 100.0% | 1.0000

B4 all | isomeric_smiles | 2 | 100.0% | 1.0000

B4 all | iupac | 2 | 100.0% | 1.0000

B4 all | moljson | 2 | 100.0% | 1.0000

B4 all | randomized_smiles | 2 | 100.0% | 1.0000

B4 all | selfies | 2 | 100.0% | 1.0000

B5 all | canonical_smiles | 2 | 100.0% | 1.0000

B5 all | cml | 2 | 100.0% | 0.5000

B5 all | deepsmiles | 2 | 100.0% | 1.0000

B5 all | inchi | 2 | 100.0% | 1.0000

B5 all | isomeric_smiles | 2 | 100.0% | 1.0000

B5 all | iupac | 2 | 100.0% | 1.0000

B5 all | moljson | 2 | 100.0% | 0.5000

B5 all | randomized_smiles | 2 | 100.0% | 0.5000

B5 all | selfies | 2 | 100.0% | 0.0000

B6 all | canonical_smiles | 2 | 100.0% | 0.5000

B6 all | cml | 2 | 100.0% | 0.0000

B6 all | deepsmiles | 2 | 100.0% | 0.0000

B6 all | inchi | 2 | 100.0% | 0.0000

B6 all | isomeric_smiles | 2 | 100.0% | 0.0000

B6 all | iupac | 2 | 100.0% | 0.0000

B6 all | moljson | 2 | 100.0% | 0.5000

B6 all | randomized_smiles | 2 | 100.0% | 0.5000

B6 all | selfies | 2 | 100.0% | 0.0000

B9 all | canonical_smiles | 2 | 100.0% | 0.5000

B9 all | cml | 2 | 100.0% | 0.5000

B9 all | deepsmiles | 2 | 100.0% | 1.0000

B9 all | inchi | 2 | 100.0% | 0.5000

B9 all | isomeric_smiles | 2 | 100.0% | 0.5000

B9 all | iupac | 2 | 100.0% | 0.0000

B9 all | moljson | 2 | 100.0% | 0.5000

B9 all | randomized_smiles | 2 | 100.0% | 0.5000

B9 all | selfies | 2 | 100.0% | 1.0000

B10 all | canonical_smiles | 2 | 100.0% | 1.0000

B10 all | cml | 2 | 100.0% | 1.0000

B10 all | deepsmiles | 2 | 100.0% | 1.0000

B10 all | inchi | 2 | 100.0% | 1.0000

B10 all | isomeric_smiles | 2 | 100.0% | 1.0000

B10 all | iupac | 2 | 100.0% | 0.0000

B10 all | moljson | 2 | 100.0% | 1.0000

B10 all | randomized_smiles | 2 | 100.0% | 1.0000

B10 all | selfies | 2 | 100.0% | 1.0000